



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to K.K FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

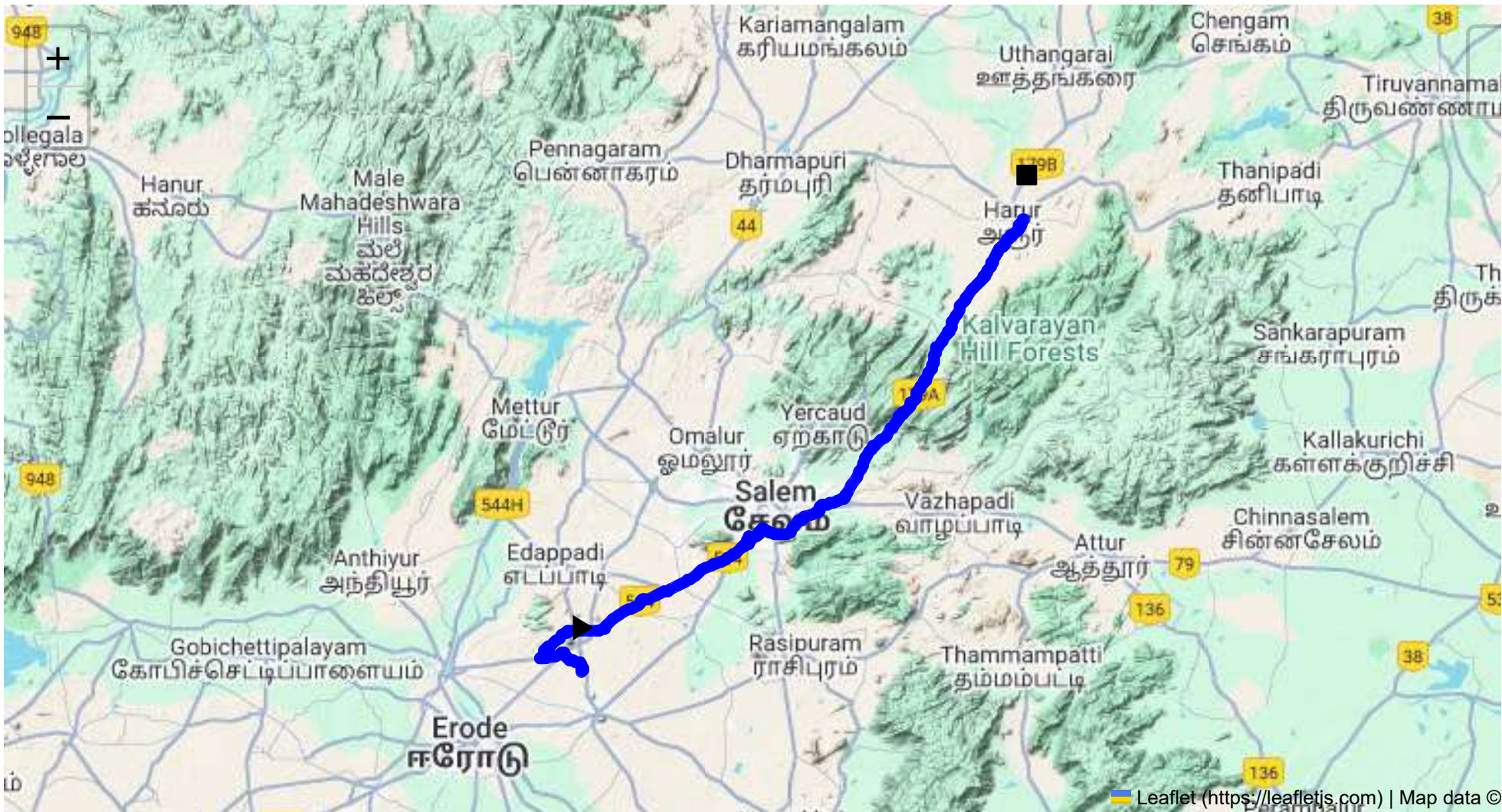
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 117.67 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.056437, 78.49112)



Welcome to the Journey Risk Management Study

Route Overview: The route from CVQF+23W in Sangagiri to 3F4R+J6J in Harur covers approximately 117.67 kilometers. It includes several major roads such as the Sankari RS - Erode Privu Rd, Bhavani Main Road, and NH544, passing through urban and rural areas, including key towns like Sankari and Salem.

1. Overview of the Route Map: This route involves a mix of National Highways and state roads. It begins in the industrial region of Sangagiri, travels north through Sankari and Padaiveedu, and passes through the

city of Salem before continuing east to Harur. The route is defined by a mix of urban congestion and more open rural stretches.

2. Typical Weather Conditions and Hazards: Tamil Nadu typically has a tropical climate, with hot summers and monsoon rains from June to September. During the monsoon, heavy rains can cause flooding and reduced visibility. Summers can be extremely hot, which might affect vehicle performance, particularly cooling systems in heavy vehicles.

3. Traffic Patterns: Traffic congestion is notable in urban areas such as Salem. Peak hours usually occur between 8-10 AM and 5-8 PM. During these hours, delays are likely, especially on NH544 and major intersections in Salem. Rural segments are generally less congested.

4. Road Quality and Infrastructure: The national highways (e.g., NH544) are well-maintained with multiple lanes, but certain sections near towns may have narrower roads with potential potholes. The state roads are generally narrower and may not always be in optimal condition, particularly post-monsoon.

5. Alternative Routes for Emergencies: In emergencies, alternative routes via NH44 offer detours around congested areas in Salem. For those in the western part of the route near Erode, bypass roads can help circumvent urban traffic jams.

6. Local Regulations for Hazardous Materials: Transport of hazardous materials requires compliance with regulations specific to Tamil Nadu, such as proper labeling, documentation, and restrictions on transit through densely populated areas during peak hours.

7. Historical Incidents: There have been occasional incidents involving heavy vehicles in Salem, primarily due to sharp turns and heavy traffic. The NH544 section has had reports of accidents due to overspeeding and poor visibility during heavy rains.

8. Environmental Considerations: The route traverses agricultural areas. Drivers should be cautious near sensitive zones, such as water bodies and wildlife areas, as leakage of hazardous materials could cause significant ecological damage.

9. Communication Coverage: Most of the route has good cellular network coverage, particularly in urban areas. However, certain rural segments might have spotty connectivity, posing challenges for timely communication.

10. Emergency Response Times: In urban areas like Salem, emergency response is typically quick, within 15-30 minutes. Rural sections may experience longer response times, up to an hour, due to distance and road accessibility.

12. Overall Risk Assessment: This route is moderately risky due to weather-related hazards, urban congestion, and challenging road conditions in sections. Preparing for variable conditions, avoiding peak hours, and ensuring good vehicle maintenance can mitigate potential issues.

The trip requires careful planning and alertness, particularly with regards to regulatory compliance and environmental protection.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45571, 77.81702	30 KM/Hr
9	Turn	High	11.45615, 77.81677	15 KM/Hr
10	Turn	Medium	11.53560, 77.95773	30 KM/Hr
11	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
12	Turn	High	11.62130, 78.10776	15 KM/Hr
13	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
14	Turn	Medium	11.61224, 78.10469	30 KM/Hr
15	Turn	Medium	11.62537, 78.12040	30 KM/Hr
16	Turn	Medium	11.63283, 78.12757	30 KM/Hr
17	Turn	Medium	11.62901, 78.16571	30 KM/Hr
18	Turn	Medium	11.65803, 78.20452	30 KM/Hr
19	Turn	Medium	11.66354, 78.20645	30 KM/Hr
20	Blind Spot	Blind Spot	11.67120, 78.23136	10 KM/Hr
21	Turn	High	11.67171, 78.23139	15 KM/Hr
22	Turn	Medium	11.79052, 78.32170	30 KM/Hr
23	Turn	Medium	11.80458, 78.33335	30 KM/Hr
24	Turn	Medium	11.81249, 78.34088	30 KM/Hr
25	Turn	Medium	11.83492, 78.35299	30 KM/Hr
26	Turn	Medium	11.84761, 78.36080	30 KM/Hr
27	Turn	Medium	11.99572, 78.44559	30 KM/Hr
28	Turn	Medium	12.01595, 78.45693	30 KM/Hr
29	Turn	Medium	12.04269, 78.48340	30 KM/Hr
30	Turn	High	12.04666, 78.48405	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
31	Turn	Medium	12.04646, 78.48498	30 KM/Hr
32	Turn	Medium	12.04663, 78.48554	30 KM/Hr
33	Turn	Medium	12.04722, 78.48607	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
19	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
20	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
21	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
22	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
23	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
24	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
25	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
28	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
30	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
33	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
26	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
27	school	Kailash Manasarover School	11.6528379, 78.2021579	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
31	school	Indian Matriculation Higher secondary school	12.006299, 78.4531204	30 km/h	Medium
32	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
34	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
35	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
36	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
37	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
38	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53560, 77.95773



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.62145, 78.10740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.61235, 78.10438



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61224, 78.10469



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62537, 78.12040



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63283, 78.12757



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62901, 78.16571



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65803, 78.20452



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66354, 78.20645



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67171, 78.23139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.79052, 78.32170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80458, 78.33335



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.81249, 78.34088



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83492, 78.35299



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84761, 78.36080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99572, 78.44559



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01595, 78.45693



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04269, 78.48340



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04666, 78.48405



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04646, 78.48498



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04663, 78.48554



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04722, 78.48607

